PROJECT COLLECTION

REZA BABAPIRI

Mechanical Design & Manufacturing

CAD, industrial equipment and digital manufacturing

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2 TECHNICAL CASE STUDIES

Engineering scope / Method / Results / Visual evidence / Limitations

Technical design, coding and analysis: Reza Babapiri

reza-babapiri.github.io

PROFESSIONAL EXPERIENCE · INDUSTRIAL EQUIPMENT

Industrial Expansion Joints | Design Practice

Bellows, flanges, piping components and mechanical design verification

ENGINEERING QUESTION

How can expansion joints and connected piping components be sized and selected for pressure, displacement, corrosion exposure and fabrication requirements?

ROLE & CONTRIBUTION

Mechanical Design Engineer at Enetaf Sane Co. I performed industrial design, technical review, load-related checks, materials work and drawing preparation. Employer and client drawings are intentionally not reproduced.

ENGINEERING DECISIONS

- Translated operating-service requirements into bellows, expansion-joint, flange and piping-component configurations.
- Reviewed guide-rod arrangement, bolt quantity and load paths, and revised flange thickness according to applied mechanical loads.
- Selected or revised materials and wall thickness where corrosion exposure or specified service conditions required changes.
- Assessed structural response in SolidWorks Simulation and ABAQUS; coordinated design feedback with fabrication colleagues.

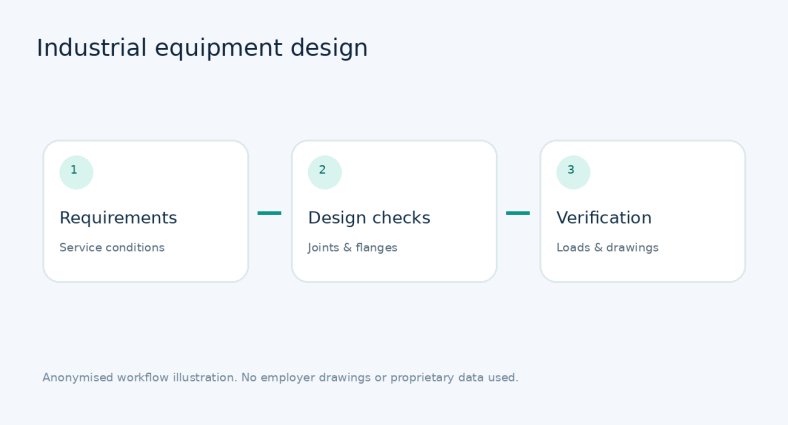
SELECTED ENGINEERING EVIDENCE

Engineering choices were checked with EJMA-based calculation software, applicable ASME / project requirements and structural calculations. No public numerical or certified design result is claimed.

VISUAL EVIDENCE · PROFESSIONAL EXPERIENCE · INDUSTRIAL EQUIPMENT

Industrial Expansion Joints | Design Practice

Original images, models and results from the supplied project material



Schematic presentation of expansion-joint design topics; illustration only, no company-owned drawings.

RESULTS & ENGINEERING INTERPRETATION

Engineering choices were checked with EJMA-based calculation software, applicable ASME / project requirements and structural calculations. No public numerical or certified design result is claimed.

DIGITAL MANUFACTURING · CAD/CAM

Fusion 360 | CAD/CAM CNC Workflow

Aluminium 6061 milling plan with machining simulation and NC output

ENGINEERING QUESTION

How can a parametrically modelled aluminium part be translated into a verifiable two-dimensional CNC machining workflow?

ROLE & CONTRIBUTION

Independently prepared the part geometry, material selection, machining setup, toolpath sequence, simulation and NC output for the course challenge.

ENGINEERING DECISIONS

- Modelled the feature geometry in Fusion 360; assigned Aluminium 6061 as the workpiece material.
- Defined setup direction, work coordinate origin and stock definition to align programming with the intended fixture orientation.
- Sequenced Face, 2D Adaptive, Pocket, Contour and Drill operations rather than relying on one generic toolpath.
- Simulated material removal and used a generic 2D post-processor to export an NC file.

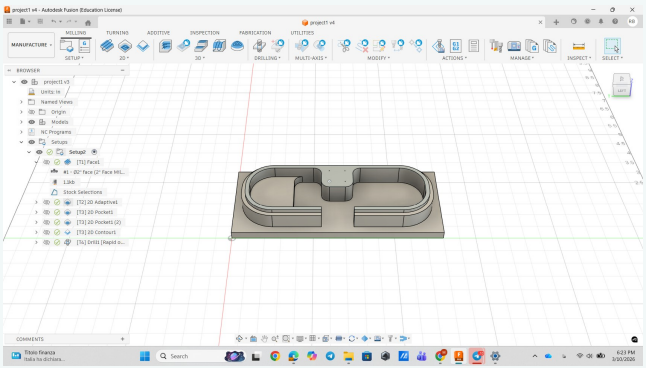
SELECTED ENGINEERING EVIDENCE

The supplied Fusion screenshots show the model, material assignment, setup parameters and toolpath simulation. The course report records complete simulated stock removal and exported NC output.

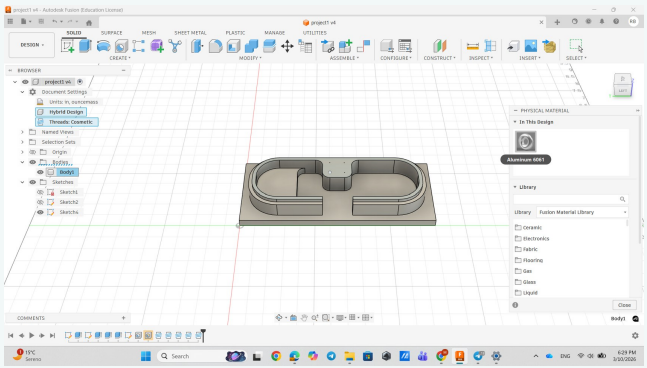
VISUAL EVIDENCE · DIGITAL MANUFACTURING · CAD/CAM

Fusion 360 | CAD/CAM CNC Workflow

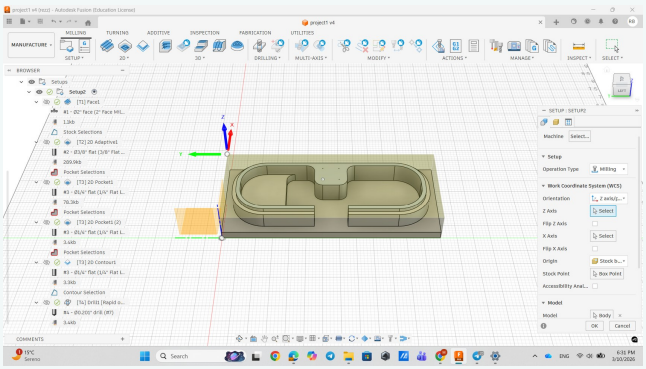
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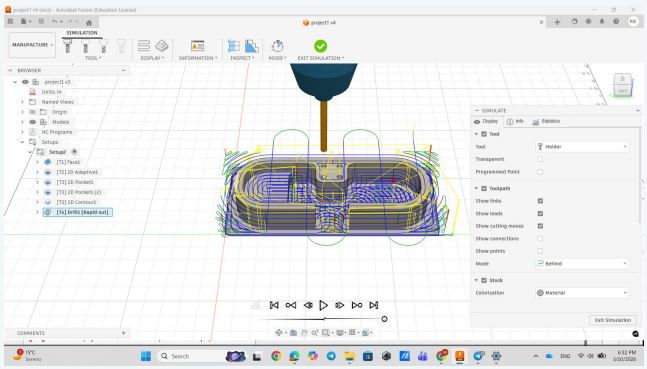
Original Fusion 360 CAD geometry showing the part, pockets and machined features.



Fusion 360 physical material assigned to the model: Aluminium 6061.



Manufacturing workspace showing work coordinate system and stock origin setup.



CAM simulation of 2D milling and drilling paths, including leads and linking moves.

RESULTS & ENGINEERING INTERPRETATION

The supplied Fusion screenshots show the model, material assignment, setup parameters and toolpath simulation. The course report records complete simulated stock removal and exported NC output.